COMPUTER NETWORKS II

ΠΑΝΕΠΙΣΤΗΜΙΟ ΔΥΤΙΚΗΣ ΑΤΤΙΚΗΣ
UNIVERSITY OF WEST ATTICA

DEPARTMENT OF INFORMATION AND COMPUTER ENGINEERING

1
OSPF ROUTING

STUDENT DETAILS

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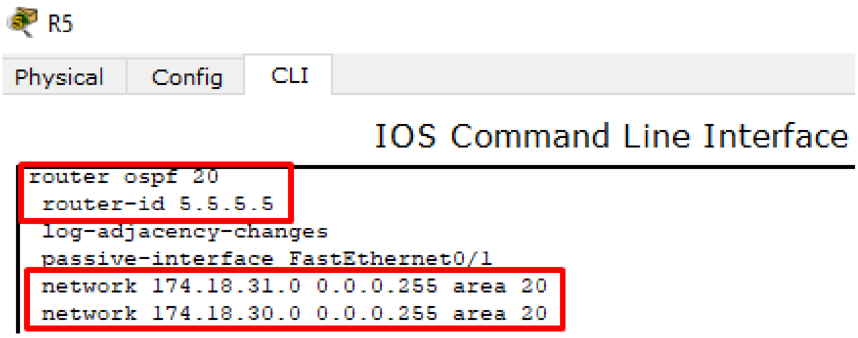
Question 1

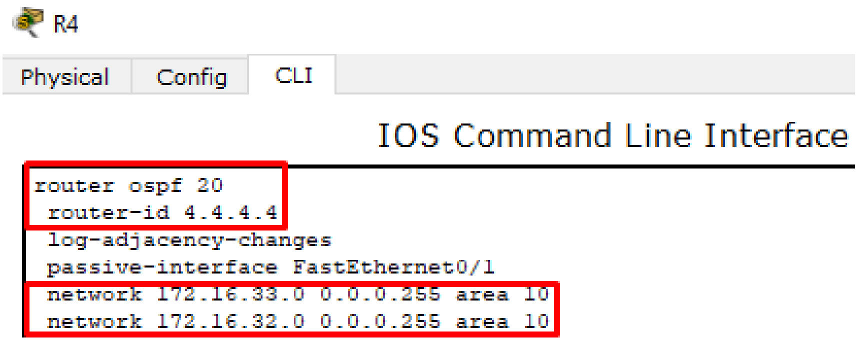
Make the appropriate settings on the routers so that there is connectivity between all the network devices of the Autonomous System 20. To route the traffic between the routers, you should make use of the OSPF protocol. Specifically, the configuration of the OSPF protocol in a router concerns:
§ The configuration of an OSPF process in the router.
§ Defining a unique router identifier (optional step).
§ The definition of the subnets connected to the router and the region (area) to which they belong.

To make the appropriate settings on the routers so that there is connectivity between all network devices of the Autonomous System 20, the OSPF protocol is used . To configure an OSPF process on the router, follow these steps:

1. We enter the " CLI " of the router.
2. With the " enable " command we enter " privileged mode » of the router.
3. We execute the command " router ospf < process - id >" to enter " router configuration mode " of the router, where " process - id " can be any integer for routers belonging to the same autonomous system (eg for router " R 4", " router ospf 20").
4. We define with the command " router - id < OSPF process id in ip address format >" a unique identifier of the router, where it will be in " ip " address format (eg for router " R 4", " router - id 4.4.4.4").
5. With the command " network < ip - address > < wildcard - mask > area < area - id >" we define the subnets connected to the router and the region to which they belong (e.g. to define a subnet connected to the router " R 4" and the region " area 10" to which it belongs, we execute the command " network 172.16.32.0 0.0.0.255 area 10"). The wildcard - mask is the subnet mask that interfaces to the corresponding router with the aces reversed to zeros and vice versa.

With the command " show run » to « privileged mode " in the " CLI " of each router (R 1, ABR 2, ABR 3, R 4, R 5) we see the summary results of running the commands "3, 4, 5" for each subnet connected to each router and the region it belongs to.





R1

Physical | Config | CLI

IOS Command Line Interface

```
!
router ospf 20
 router-id 1.1.1.1
 log-adjacency-changes
 network 173.17.37.0 0.0.0.255 area 0
 network 173.17.38.0 0.0.0.255 area 0
```

ABR2

Physical | Config | CLI

IOS Command Line Interface

```
!
router ospf 20
 router-id 2.2.2.2
 log-adjacency-changes
 area 0 range 173.17.36.0 255.255.252.0
 area 10 range 172.16.32.0 255.255.254.0
 network 172.16.32.0 0.0.0.255 area 10
 network 173.17.37.0 0.0.0.255 area 0
 network 173.17.39.0 0.0.0.255 area 0
```

ABR3

Physical | Config | CLI

IOS Command Line Interface

```
!
router ospf 20
 router-id 3.3.3.3
 log-adjacency-changes
 area 0 range 173.17.36.0 255.255.252.0
 area 20 range 174.18.30.0 255.255.254.0
 network 173.17.38.0 0.0.0.255 area 0
 network 173.17.39.0 0.0.0.255 area 0
 network 174.18.30.0 0.0.0.255 area 20
```

With the command " show ip route » to « privileged mode " in the " CLI " of each router (R 1, ABR 2, ABR 3, R 4, R 5, Router) we see the routing table that contains information about the subnets interconnected with the corresponding router and with which protocol the communication is achieved them and from which interface.

R4

Physical Config CLI

IOS Command Line Interface

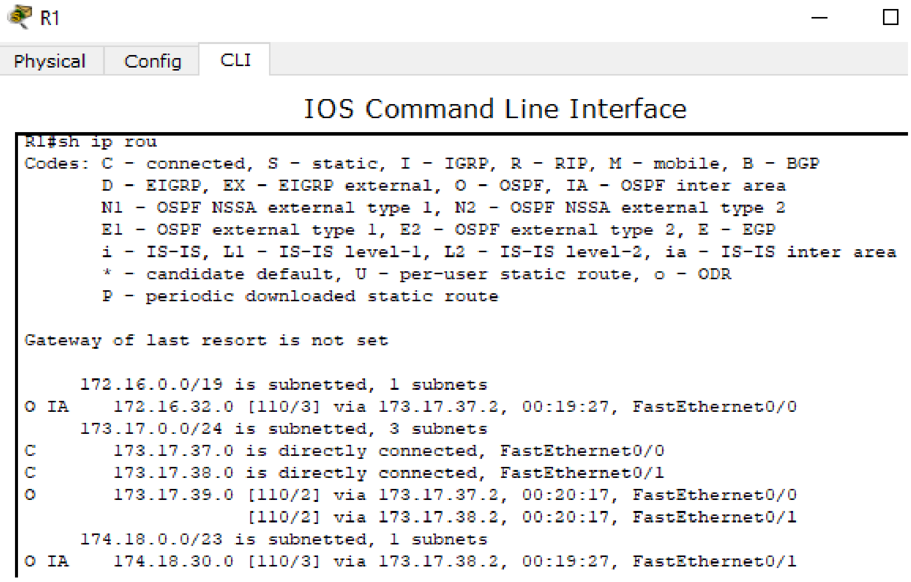
```
R4#sh ip rou
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.16.0.0/24 is subnetted, 3 subnets
C       172.16.32.0 is directly connected, FastEthernet0/0
C       172.16.33.0 is directly connected, FastEthernet0/1
C       172.16.34.0 is directly connected, Serial0/0/0
     173.17.0.0/22 is subnetted, 1 subnets
O IA    173.17.36.0 [110/2] via 172.16.32.1, 05:11:48, FastEthernet0/0
     174.18.0.0/23 is subnetted, 1 subnets
O IA    174.18.30.0 [110/4] via 172.16.32.1, 01:39:33, FastEthernet0/0
```

R5

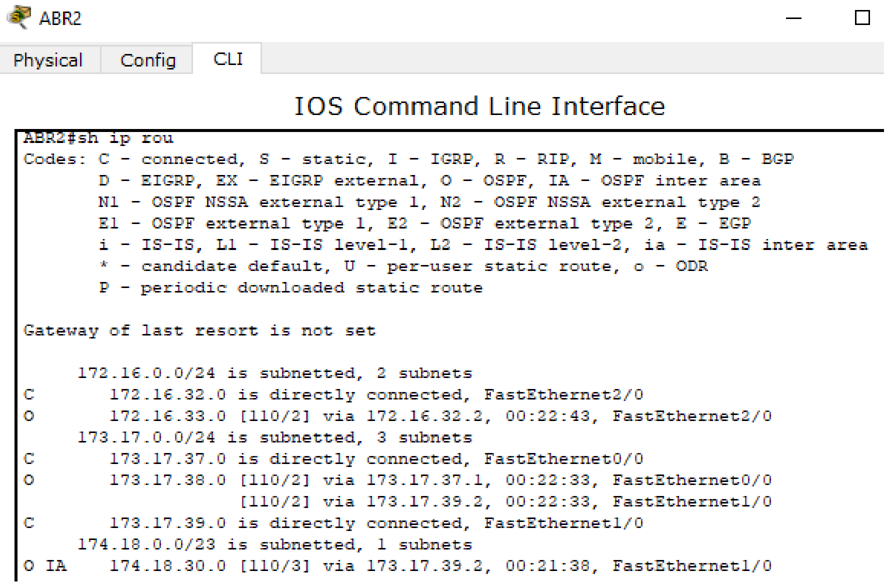
Physical Config CLI

IOS Command Line Interface

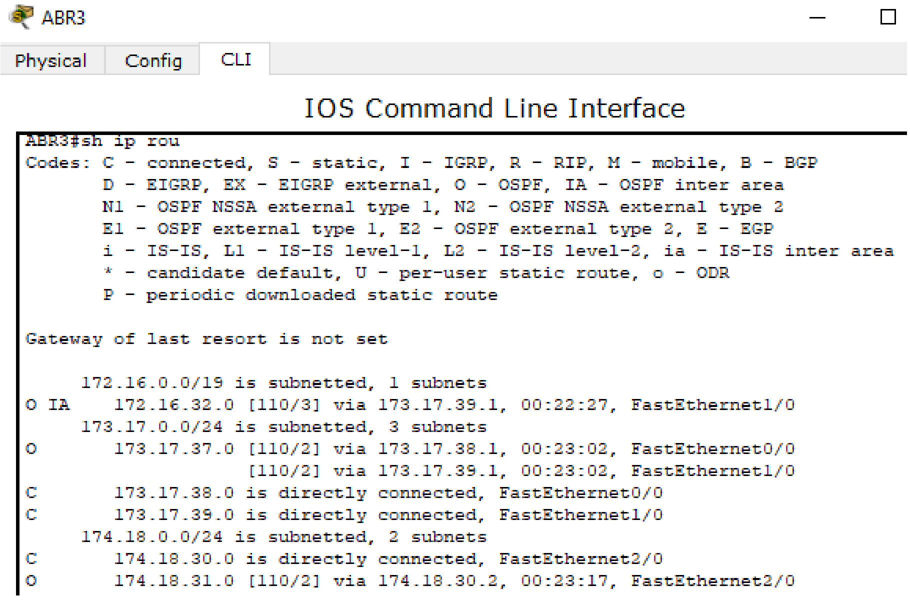
```
R5#sh ip rou
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.16.0.0/19 is subnetted, 1 subnets
O IA    172.16.32.0 [110/4] via 174.18.30.1, 00:18:14, FastEthernet0/0
     173.17.0.0/22 is subnetted, 1 subnets
O IA    173.17.36.0 [110/2] via 174.18.30.1, 00:19:04, FastEthernet0/0
     174.18.0.0/24 is subnetted, 2 subnets
C       174.18.30.0 is directly connected, FastEthernet0/0
C       174.18.31.0 is directly connected, FastEthernet0/1
```

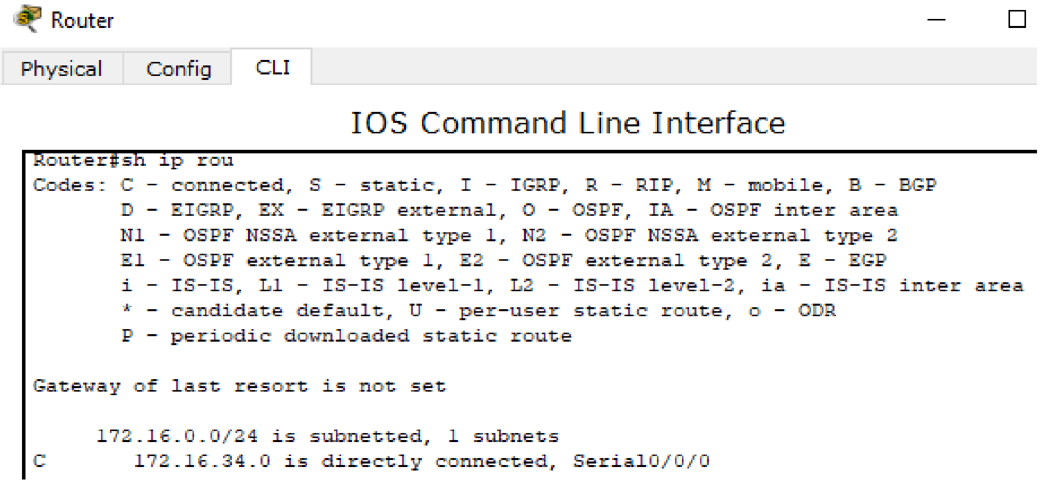



```
R1#sh ip rou
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.16.0.0/19 is subnetted, 1 subnets
O IA    172.16.32.0 [110/3] via 173.17.37.2, 00:19:27, FastEthernet0/0
     173.17.0.0/24 is subnetted, 3 subnets
C       173.17.37.0 is directly connected, FastEthernet0/0
C       173.17.38.0 is directly connected, FastEthernet0/1
O       173.17.39.0 [110/2] via 173.17.37.2, 00:20:17, FastEthernet0/0
                    [110/2] via 173.17.38.2, 00:20:17, FastEthernet0/1
     174.18.0.0/23 is subnetted, 1 subnets
O IA    174.18.30.0 [110/3] via 173.17.38.2, 00:19:27, FastEthernet0/1
```




```
ABR2#sh ip rou
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.16.0.0/24 is subnetted, 2 subnets
C       172.16.32.0 is directly connected, FastEthernet2/0
O       172.16.33.0 [110/2] via 172.16.32.2, 00:22:43, FastEthernet2/0
     173.17.0.0/24 is subnetted, 3 subnets
C       173.17.37.0 is directly connected, FastEthernet0/0
O       173.17.38.0 [110/2] via 173.17.37.1, 00:22:33, FastEthernet0/0
                    [110/2] via 173.17.39.2, 00:22:33, FastEthernet1/0
C       173.17.39.0 is directly connected, FastEthernet1/0
     174.18.0.0/23 is subnetted, 1 subnets
O IA    174.18.30.0 [110/3] via 173.17.39.2, 00:21:38, FastEthernet1/0
```

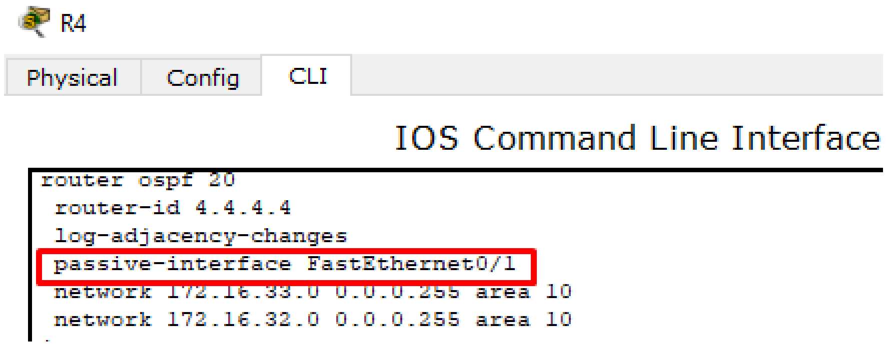


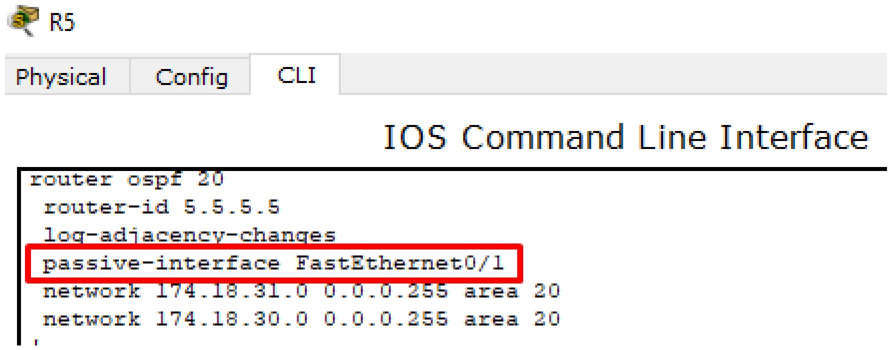


Question 2

Set the f0/1 interfaces of routers R4 and R5 as passive so that no OSPF updates are sent from these interfaces.

In order to set the " f 0/1" interfaces of routers " R 4" and " R 5" as passive , so that no " OSPF " updates are sent from these interfaces, we enter " router configuration mode " and execute the command " passive - interface f 0/1”.
With the command " show run » to « privileged mode ” in the “ CLI ” of each router (R 4, R 5) we see the summary results of running the command for which interface of the respective router is set as passive.



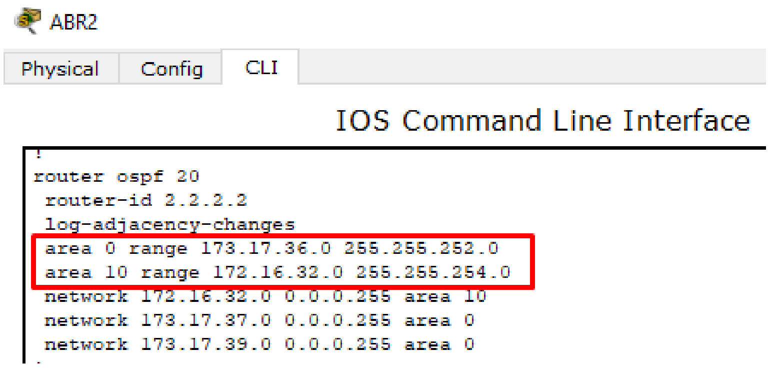


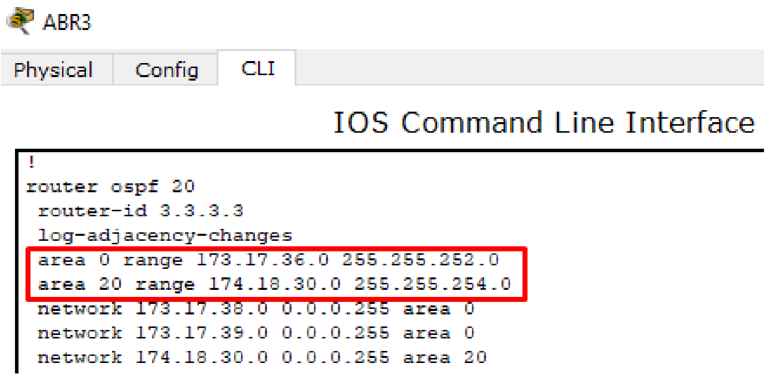
Question 3

Configure the Area Border Routers of the Autonomous System 20 to summarize the routes of the regions that are interconnected (inter-area summarization).

To configure the “ Area Border Routers » of the Autonomous System 20, so as to summarize the routes of the regions that are interconnected (inter - area summarization), we enter " router configuration mode " and execute the command " area < area - id > range < IP address > < mask >. The " Area Border Routers " are "AB R 2" and "AB R 3" and their purpose is to advertise the " summarization " of the routes of each region (area 0, area 10, area 20).

With the command " show run » to « privileged mode " in the " CLI " of each router (AB R 2, AB R 3) we see the summary results of the " summarization " of the routes advertised by the " Area Border Routers '.





In " area 10" we have the following subnets with their respective " prefixes ":

– 172.16.32.0/24
– 172.16.33.0/24

Each " IP " address in the binary system is respectively:

– 10101100.00010000.0010000 0.00000000
– 10101100.00010000.0010000 1.00000000

The initial " prefix " of the networks to be added is 24 and our goal is to get a smaller one, so that we finally have an " IP " network address keeping the total original address range of the two subnets. The first 23 bits of the two subnets are common and the 1 bit that will be returned to the " host portion " alternates in combinations of 0 and 1. Therefore, the " IP " address of the " summarized " network will be the common bits of the two networks with the bits of the " host portion " to be all converted to 0, that is, the address 172.16.32.0 and the mask will be the number of **common bits** of the " IP " addresses that we want to " summarize ", that is, /23.

In " area 20" we have the following subnets with their respective " prefixes ":

– 174.18.30.0/24
– 174.18.31.0/24

Each " IP " address in the binary system is respectively:

– 101011 1 0.000100 1 0. 0001111 0 .00000000
– 101011 1 0.000100 1 0.00 01111 1 .00000000

The initial " prefix " of the networks to be added is 24 and our goal is to get a smaller one, so that we finally have an " IP " network address keeping the total original address range of the two subnets. The first 23 bits of the two subnets are common and the 1 bit that will be returned to the " host portion " alternates in combinations of 0 and 1. Therefore, the " IP " address of the " summarized " network will be the common bits of the two networks with the bits of the " host portion " to be all converted to 0, that is, the address 174.18.30.0 and the mask will be the number of **common bits** of the " IP " addresses that we want to " summarize ", that is, /23.

In " area 0" we have the following subnets with their respective " prefixes ":

– 173.17.37.0/24
– 173.17.38.0/24
– 173.17.39.0/24

Each " IP " address in the binary system is respectively:

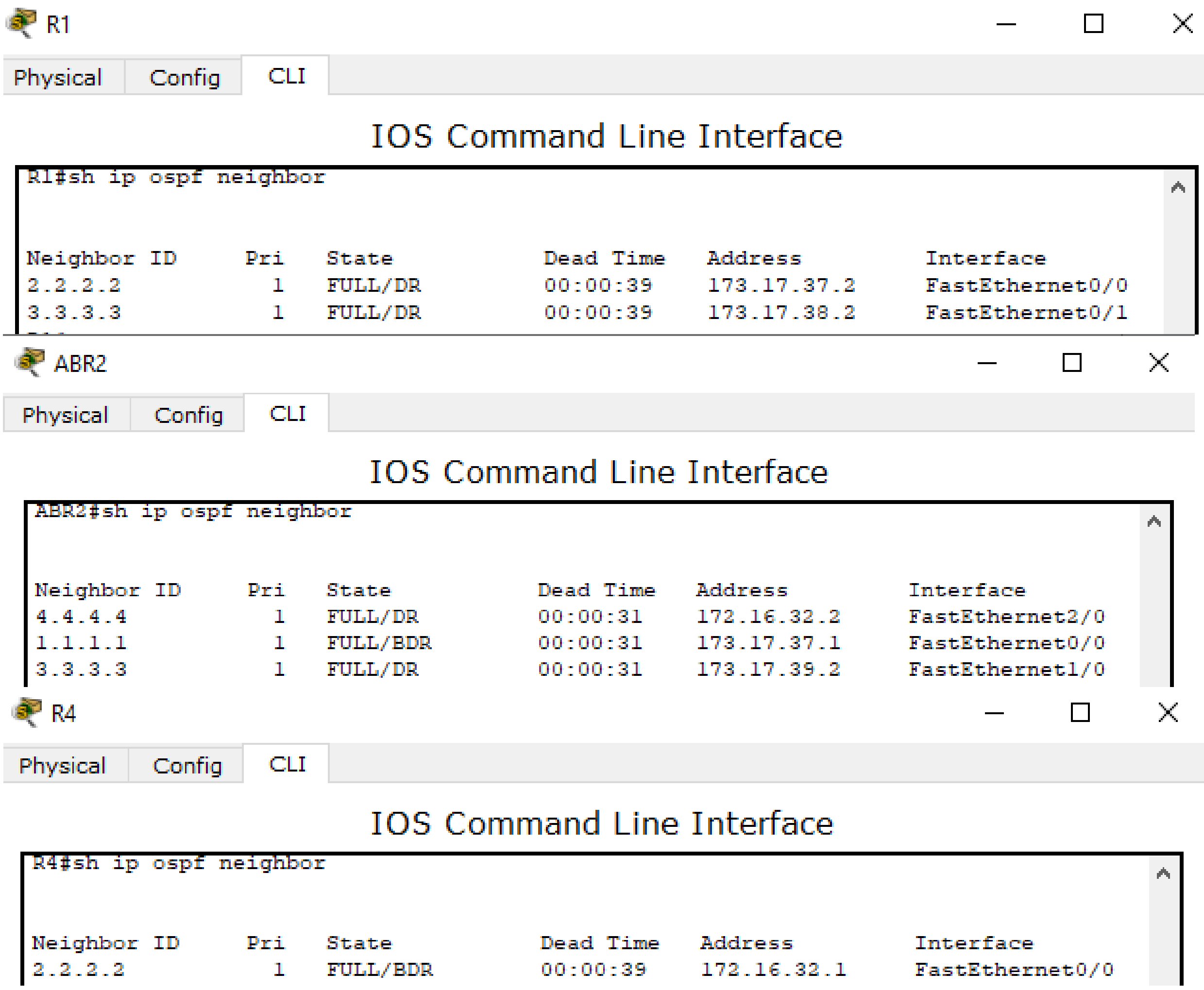
- 1010110 1 .0001000 1 . 001001 01 .00000000
- 1010110 1 .0001000 1 . 001001 10 .00000000
- 10101101.00010001.001001 11.00000000

The initial " prefix " of the networks to be added is 24 and our goal is to get a smaller one, so that we finally have an " IP " network address keeping the total original address range of the two subnets. The first 22 bits of the three subnets are common and the 2 bits that will be returned to the " host portion " alternates in the combinations 01, 10 and 11. Therefore, the " IP " address of the " summarized " network will be the common bits of the three networks with the bits of the " host portion " to be all converted to 0, that is, the address 173.16.36.0 and the mask will be the number of **common bits** of the " IP " addresses that we want to " summarize ", that is, /22.

Question 4

Study the neighbor states of routers R1, R2, and R4.

To study neighborhood relations (neighboring states) of routers " R 1, ABR 2, R 4", we enter " privileged mode " and execute the command " show ip ospf neighbor '.



The table presented by the " show ip ospf neighbor » has the following features:

1. Neighbor ID : The unique identifier of the neighboring router that the router's interconnection with the neighbor has been made with the “ OSPF ” protocol.
2. Pri : The priority number of the neighbor to receive " OSPF " updates.
3. Country : The connection status of the neighbor router (eg " FULL ", ie, the neighbor's " OSPF " settings are discoverable for advertisement).
4. Address : The " ip " address of the interface of the neighboring router connected to the corresponding router.
5. Interface : The neighbor's interface with the router.

From the above tables we conclude that the neighbor relations of each of the three routers (R 1, ABR 2, R 4) are as follows:

Neighbors R 1 (ABR 2, ABR 3)

From the table of " R 1" we find that it is contained in the column " Neighbor ID » the unique identifier of the router « ABR 2 » (2.2.2.2). In the " State " column we notice that the connection status of the neighboring routers is " FULL / DR ". In other words, “ R 1 ” has neighbor relationships with both “ Area Border Routers " (in the column " Neighbor ID " also contains the unique identifier of the router " ABR 3" which is 3.3.3.3) and since it is " Designated Routers ” and in “ FULL ” connection state , “ R 1” will exchange “ OSPF ” updates only with them .

ABR 2 neighbors (R 1, ABR 3, R 4)

From the table of " ABR 2" we find that it is contained in the column " Neighbor ID » the unique identifier of the router « R 1 » (1.1.1.1). In the " State " column we notice that the connection status of two neighboring routers is " FULL / DR " and another one is " FULL / BDR ". More specifically, " ABR 2" has neighboring relations with " Area Border Router » " ABR 3" and with the internal router of area 10 " R 4" (in the column " Neighbor ID " contains both the unique identifier of router " ABR 3" which is 3.3.3.3 and router " R 4" which is 4.4.4.4). " ABR 3" and " R 4" are in connection status " FULL " and since they are " Designated Routers , ABR 2 will exchange OSPF updates only with them. " R 1 " is " Backup Designated Router " and in connection status " FULL ", which means that in case it replaces the " Designated Routers ", he will receive all " OSPF " updates.

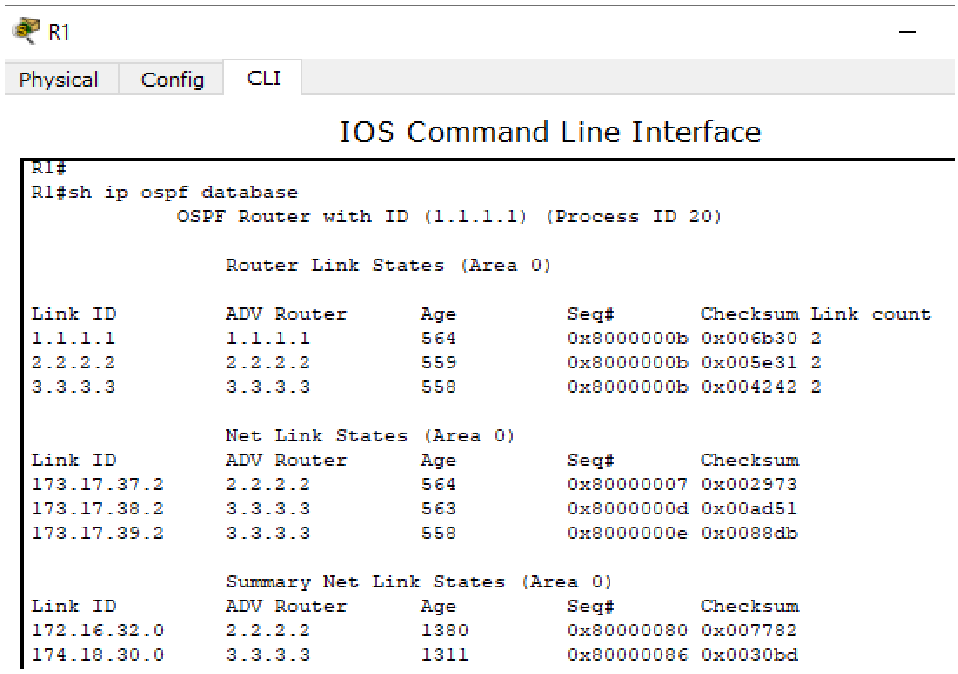
Neighbors R 4 (ABR 2)

From the table of " R 4" we find that it is contained in the column " Neighbor ID ” the unique identifier of router “ ABR 2” (2.2.2.2). In the " State " column we notice that the connection status of the neighboring router is " FULL / BDR ". In other words, " ABR 2 " is " B ackup Designated Router ' and in ' FULL ' connection state , ' R 4' will exchange ' OSPF ' updates only with it.

Question 5

Study the OSPF Link State Database of routers R1, R2, and R4 and describe the types of LSAs they contain.

With the command " show ip ospf database » to « privileged mode " of each router, we display the " OSPF Link State Database " of the corresponding router, where it contains detailed types of " LSA " packets.

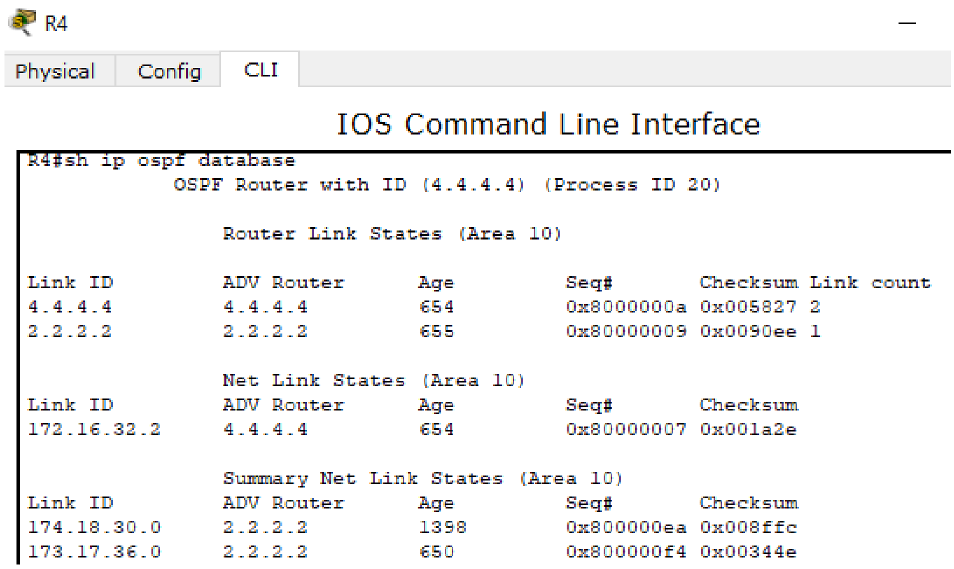


R1

Physical Config CLI

IOS Command Line Interface

```
R1#
R1#sh ip ospf database
            OSPF Router with ID (1.1.1.1) (Process ID 20)

                Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum Link count
1.1.1.1         1.1.1.1         564         0x8000000b 0x006b30 2
2.2.2.2         2.2.2.2         559         0x8000000b 0x005e31 2
3.3.3.3         3.3.3.3         558         0x8000000b 0x004242 2

                Net Link States (Area 0)
Link ID         ADV Router      Age         Seq#       Checksum
173.17.37.2     2.2.2.2         564         0x80000007 0x002973
173.17.38.2     3.3.3.3         563         0x8000000d 0x00ad51
173.17.39.2     3.3.3.3         558         0x8000000e 0x0088db

                Summary Net Link States (Area 0)
Link ID         ADV Router      Age         Seq#       Checksum
172.16.32.0     2.2.2.2         1380        0x80000080 0x007782
174.18.30.0     3.3.3.3         1311        0x80000086 0x0030bd
```



R4

Physical Config CLI

IOS Command Line Interface

```
R4#sh ip ospf database
            OSPF Router with ID (4.4.4.4) (Process ID 20)

                Router Link States (Area 10)

Link ID         ADV Router      Age         Seq#       Checksum Link count
4.4.4.4         4.4.4.4         654         0x8000000a 0x005827 2
2.2.2.2         2.2.2.2         655         0x80000009 0x0090ee 1

                Net Link States (Area 10)
Link ID         ADV Router      Age         Seq#       Checksum
172.16.32.2     4.4.4.4         654         0x80000007 0x001a2e

                Summary Net Link States (Area 10)
Link ID         ADV Router      Age         Seq#       Checksum
174.18.30.0     2.2.2.2         1398        0x800000ea 0x008ffc
173.17.36.0     2.2.2.2         650         0x800000f4 0x00344e
```

ABR2

Physical | Config | CLI

IOS Command Line Interface

```
ABR2#sh ip ospf database
            OSPF Router with ID (2.2.2.2) (Process ID 20)

                Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum Link count
2.2.2.2         2.2.2.2         694         0x8000000b 0x005e31 2
1.1.1.1         1.1.1.1         700         0x8000000b 0x006b30 2
3.3.3.3         3.3.3.3         694         0x8000000b 0x004242 2

                Net Link States (Area 0)
Link ID         ADV Router      Age         Seq#       Checksum
173.17.37.2     2.2.2.2         699         0x80000007 0x002973
173.17.38.2     3.3.3.3         699         0x8000000d 0x00ad51
173.17.39.2     3.3.3.3         694         0x8000000e 0x0088db

                Summary Net Link States (Area 0)
Link ID         ADV Router      Age         Seq#       Checksum
172.16.32.0     2.2.2.2         1527        0x8000007e 0x007685
172.16.32.0     2.2.2.2         1527        0x8000007f 0x007486
172.16.32.0     2.2.2.2         1516        0x80000080 0x007782
172.16.32.0     2.2.2.2         1528        0x8000007d 0x007884
174.18.30.0     3.3.3.3         1447        0x80000086 0x0030bd

                Router Link States (Area 10)

Link ID         ADV Router      Age         Seq#       Checksum Link count
2.2.2.2         2.2.2.2         699         0x80000009 0x0090ee 1
4.4.4.4         4.4.4.4         699         0x8000000a 0x005827 2
 --More--
                Net Link States (Area 10)
Link ID         ADV Router      Age         Seq#       Checksum
172.16.32.2     4.4.4.4         699         0x80000007 0x001a2e

                Summary Net Link States (Area 10)
Link ID         ADV Router      Age         Seq#       Checksum
173.17.36.0     2.2.2.2         694         0x800000f4 0x00344e
174.18.30.0     2.2.2.2         1442        0x800000ea 0x008ffc
```

R 1

Router " R 1" is internal, located in area 0 and with neighboring routers " Area Border Routers » « ABR 2 », « ABR 3 ». Therefore, it will contain three types of " LSA " packets which are as follows:

1. Router Link States

This packet contains information about the same router (Link ID), as well as the others located in the specific area. In more detail, we have the unique identifiers of routers " R 1" (1.1.1.1), " ABR 2" (2.2.2.2) and " ABR 3" (3.3.3.3), the identifier of the area to which the three routers belong (Area 0), the unique identifiers of routers " R 1" (1.1.1.1), " ABR 2" (2.2.2.2) and " ABR 3" (3.3.3.3) that advertise their respective " LSA " packets (ADV router), the number of connections of a router in the area with another router or routers in the same area (Link count) etc.

2. Network Link States

This packet contains information about the subnets connected to the router " R 1" and the routes routed with the " OSPF " protocol for area 0. More specifically, we have the " IP " addresses of the interfaces of the neighboring routers " ABR 2" (173.17.37.2) and " ABR 3" (173.17.38.2, 173.17.39.2), which are connected to " R 1" either directly (or via interface f 0/0 of " ABR 2" with " IP " address 173.17.37.2 either via interface f 0/0 of " ABR 3" with " IP " address 173.17.38.2) or dynamically with routing protocol " OSPF " (via interface f 1/0 of " ABR 3" with " IP » address 173.17.39.2). Also, we have the unique identifiers of routers " ABR 2" (2.2.2.2) and " ABR 3" (3.3.3.3) that advertise their respective " LSA " packets (ADV router) etc.

3. Summary Network Link States

This packet contains information about the subnets of areas 20 and 10 summarized into an " IP " address by the " Area Border Routers » " ABR 2" and " ABR 3" (inter - area summarisation). In more detail, we have the " IP " addresses advertised by routers " ABR 2" (172.16.32.0, area 10) and " ABR 3" (174.18.30.0, area 20) for each area they did respectively, the summary of the routes . Also, we have the unique identifiers of routers " ABR 2" (2.2.2.2) and " ABR 3" (3.3.3.3) that advertise their respective " LSA " packets (ADV router) etc.

R 4

Router " R 4" is internal, located in area 10 and with neighboring router " Area Border Router » « ABR 2 ». Therefore, it will contain three types of " LSA " packets which are as follows:

1. Router Link States

This packet contains information about the router itself (Link ID), as well as the others located in the specific area. In more detail, we have the unique identifiers of the routers " R 4" (4.4.4.4) and " ABR 2" (2.2.2.2), the identifier of the area to which the two routers belong (Area 10), the unique identifiers of the routers " R 4" (4.4.4.4) and " ABR 2" (2.2.2.2) advertising their respective " LSA " packets (ADV router), the number of connections of a router in the area with another router or routers in the same area (Link count) etc.

2. Network Link States

This packet contains information about the subnets connected to the router " R 4" and the routes routed with the " OSPF " protocol for area 10. More specifically, we have the " IP " address of the interface connected to the neighboring router " ABR 2" (172.16.32.2). Also, we have the unique identifier of router " R 4 " (4.4.4.4) advertising the corresponding " LSA " packet (ADV router) etc.

3. Summary Network Link States

This packet contains information about the subnets of areas 20 and 0 summarized into an " IP " address by the " Area Border Routers » " ABR 2" and " ABR 3" (inter - area summarisation). In more detail, we have the " IP " addresses advertised by the routers " ABR 2" (173.17.36.0, area 0) and " ABR 3" (174.18.30.0, area 20) for each area they did respectively, the summary of the routes . Also, we have the unique identifier of router " ABR 2" (2.2.2.2) that advertises the corresponding " LSA " packets (ADV router). By connecting " ABR 2" to " ABR 3", the former advertises to " R 4" the " IP " of area 20 that " ABR 3" summarized.

ABR 2

Router " ABR 2" is " Area Border Router ”, connects areas 0 and 10 and with neighboring routers in area 10 the internal “ R 4” and in area 0 the internal “ R 1” and “ Area Border Router » « ABR 3 ». Therefore, it will contain three types of " LSA " packets for each connecting area which are as follows:

1. Router Link States (Area 0)

This packet contains information about the router itself (Link ID), as well as the others located in the specific area. In more detail, we have the unique identifiers of the routers " R 1" (1.1.1.1), " ABR 2" (2.2.2.2) and " ABR 3" (3.3.3.3), the identifier of the area to which the three routers belong (Area 0), the unique identifiers of routers " ABR 3" (3.3.3.3) and " ABR 2" (2.2.2.2) that advertise their respective " LSA " packets (ADV router), the number of connections of a router in the area with another router or routers in the same area (Link count) etc.

2. **Network Link States (Area 0)**

This packet contains information about the subnets connected to the " ABR 2" router and the routes routed with the " OSPF " protocol for area 0. More specifically, we have the " IP " addresses of the interfaces of the neighboring router " ABR 3" (173.17.38.2, 173.17.39.2), which are connected to " ABR 2 " either directly (either through interface f 1/0 of " ABR 3 " with " IP " address 173.17.39.2 or through interface f 0/ 0 of " ABR 2" with " IP " address 173.17.37.2) or dynamically with the routing protocol " OSPF " (via interface f 0/0 of " ABR 3" with " IP " address 173.17.38.2). Also, we have the unique identifiers of routers " ABR 2" (2.2.2.2) and " ABR 3" (3.3.3.3) that advertise their respective " LSA " packets (ADV router) etc.

3. **Summary Network Link States (Area 0)**

This packet contains information about the subnets of areas 20 and 10 summarized into an " IP " address by " Area Border Routers » " ABR 2" and " ABR 3" (inter - area summarisation). In more detail, we have the " IP " addresses advertised by routers " ABR 2" (172.16.32.0, area 10) and " ABR 3" (174.18.30.0, area 20) for each area they did respectively, the summary of the routes . Also, we have the unique identifier of router " ABR 2" (2.2.2.2) that advertises the corresponding " LSA " packets (ADV router) etc.

4. **Router Link States (Area 1 0)**

This packet contains information about the router itself (Link ID), as well as the others located in the specific area. In more detail, we have the unique identifiers of the routers " R 4" (4.4.4.4) and " ABR 2" (2.2.2.2), the identifier of the area to which the two routers belong (Area 10), the unique identifiers of the routers " ABR 2" (4.4.4.4) and " R 4" (4.4.4.4) advertising their respective " LSA " packets (ADV router), the number of connections of a router in the area with another router or routers in the same area (Link count) etc.

5. **Network Link States (Area 1 0)**

This packet contains information about the subnet connected to the router " ABR 2" for area 10. More specifically, we have the " IP " address of the interface that the neighboring router " R 4" connects to " ABR 2" (172.16. 32.2). Also, we have the unique identifier of router " R 4 " (4.4.4.4) advertising the corresponding " LSA " packet (ADV router) etc.

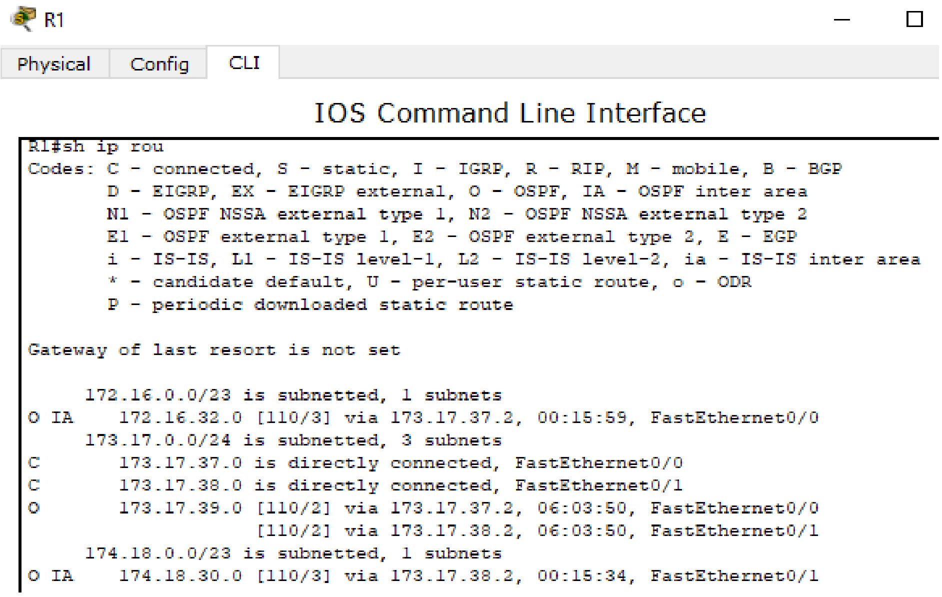
6. **Summary Network Link States (Area 10)**

This packet contains information about the subnets of areas 20 and 0 summarized into an " IP " address by the " Area Border Routers » " ABR 2" and " ABR 3" (inter - area summarisation). In more detail, we have the " IP " addresses advertised by the routers " ABR 2" (173.17.36.0, area 0) and " ABR 3" (174.18.30.0, area 20) for each area they did respectively, the summary of the routes . Also, we have the unique identifier of router " ABR 2" (2.2.2.2) that advertises the corresponding " LSA " packets (ADV router). By connecting " ABR 2" to " ABR 3", the former advertises to " R 4" the " IP " of area 20 that " ABR 3" summarized.

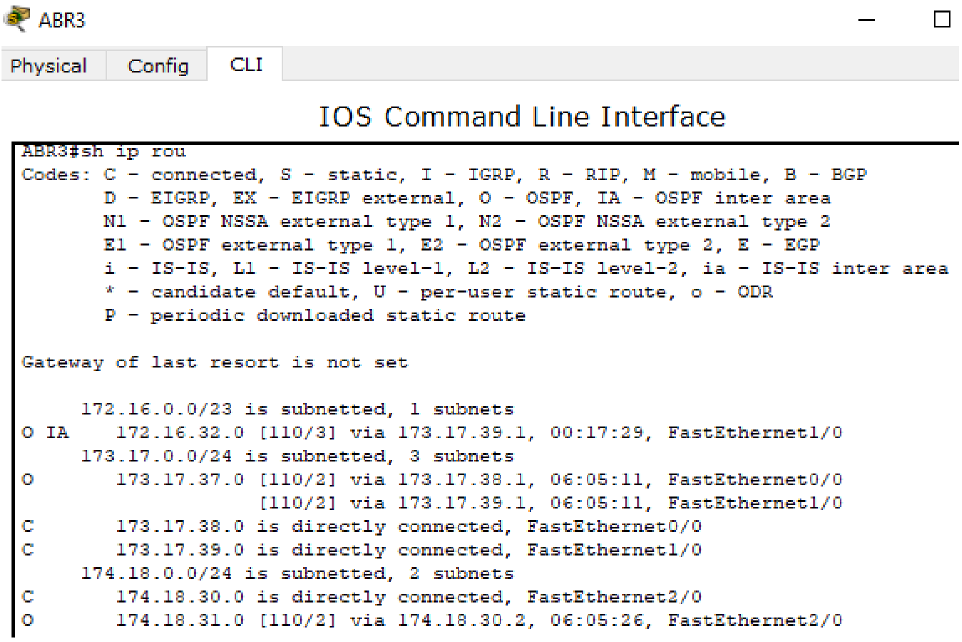
Question 6

Study the routing table of routers R1 and R3 and describe their entries

With the command " show ip route » to « privileged mode " in the " CLI " of each router (R 1, ABR 3) we see the routing table that contains information about the subnets connected to the corresponding router and with which protocol their communication is achieved and from which interface.




```
R1#sh ip rou
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.16.0.0/23 is subnetted, 1 subnets
O IA    172.16.32.0 [110/3] via 173.17.37.2, 00:15:59, FastEthernet0/0
     173.17.0.0/24 is subnetted, 3 subnets
C       173.17.37.0 is directly connected, FastEthernet0/0
C       173.17.38.0 is directly connected, FastEthernet0/1
O       173.17.39.0 [110/2] via 173.17.37.2, 06:03:50, FastEthernet0/0
                    [110/2] via 173.17.38.2, 06:03:50, FastEthernet0/1
     174.18.0.0/23 is subnetted, 1 subnets
O IA    174.18.30.0 [110/3] via 173.17.38.2, 00:15:34, FastEthernet0/1
```




```
ABR3#sh ip rou
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.16.0.0/23 is subnetted, 1 subnets
O IA    172.16.32.0 [110/3] via 173.17.39.1, 00:17:29, FastEthernet1/0
     173.17.0.0/24 is subnetted, 3 subnets
O       173.17.37.0 [110/2] via 173.17.38.1, 06:05:11, FastEthernet0/0
                    [110/2] via 173.17.39.1, 06:05:11, FastEthernet1/0
C       173.17.38.0 is directly connected, FastEthernet0/0
C       173.17.39.0 is directly connected, FastEthernet1/0
     174.18.0.0/24 is subnetted, 2 subnets
C       174.18.30.0 is directly connected, FastEthernet2/0
O       174.18.31.0 [110/2] via 174.18.30.2, 06:05:26, FastEthernet2/0
```

R 1

From the routing table of " R 1" which is an internal router, it is located in area 0 and with its neighboring routers " Area Border Routers » " ABR 2", " ABR 3" we have the following entries:

1. **C 173.17.37.0 is directly connected, FastEthernet0/0**

   The subnet "173.17.37.0/24" contains the " IP " address "173.17.37.2/24" (interface f 0/0 of " ABR 2 ") and the " IP " address "173.17.37.1/24" (interface f 0/0 of " R 1") which achieves the direct connection of " R 1" to " ABR 2".

2. **C 173.17.38.0 is directly connected, FastEthernet0/1**

The subnet "173.17.38.0/24" contains the " IP " address "173.17.38.2/24" (interface f 0/0 of " ABR 3 ") and the " IP " address "173.17.38.1/24" (interface f 0/1 of " R 1") which achieves the direct connection of " R 1" to " ABR 3".

3. **The 173.17.39.0 [110/2] via 173.17.37.2, 06:15:04, FastEthernet0/0**
   [110/2] via 173.17.38.2, 06:15:04, FastEthernet0/1

The connection of " R 1 " to the subnet " 173.17.39.0\24 " is achieved with the routing protocol " OSPF " through either interface f 0/0 (173.17.37.1/24) to interface f 0/0 of " ABR 2" (173.17.37.2/24) or via interface f 0/1 (173.17.38.1/24) to interface f 0/0 of " ABR 3" (173.17.38.2\24").

4. **O IA 172.16.32.0 [110/3] via 173.17.37.2, 00:27:12, FastEthernet0/0**

The connection of " R 1 " to the subnet " 173.17.32.0\23 ", where all the routes of area 10 have been summarized as one " IP " address from the neighbor " Area Border Router » " ABR 2" (IA) is achieved with the routing protocol " OSPF " (O IA) from interface f 0/0 (173.17.37.1/24) to interface f 0/0 of " ABR 2" (173.17.37.2/24).

5. **O IA 174.18.30.0 [110/3] via 173.17.38.2, 00:13:33, FastEthernet0/1**

The connection of " R 1 " to the subnet " 174.18.30.0\23 ", where all the routes of area 20 have been summarized as one " IP " address from the neighbor " Area Border Router » " ABR 3" (IA) is achieved with the routing protocol " OSPF " (O IA) from interface f 0/1 (173.17.38.1/24) to interface f 0/0 of " ABR 3" (173.17.38.2/24).

ASBR3

1. **C 173.17.38.0 is directly connected, FastEthernet0/0**

The subnet "173.17.38.0/24" contains the " IP " address "173.17.38.2/24" (interface f 0/0 of " ABR 3 ") and the " IP " address "173.17.38.1/24" (interface f 0/0 of " R 1") which achieves the direct connection of " R 1" to " ABR 3".

2. **C 173.17.39.0 is directly connected, FastEthernet1/0**

The subnet "173.17.39.0/24" contains the " IP " address "173.17.39.2/24" (interface f 1/0 of " ABR 3 ") and the " IP " address "173.17.39.1/24" (interface f 1/0 of " ASBR 2") which achieves the direct connection of " ABR 2" to " ABR 3".

3. **C 174.18.30.0 is directly connected, FastEthernet2/0**

The subnet "174.18.30.0/24" contains the " IP " address "174.18.30.1/24" (interface f 2/0 of " ABR 3 ") and the " IP " address "174.18.30.2/24" (interface f 0/0 of " R 5") which achieves the direct connection of " R 5" to " ABR 3".

4. **O 173.17.37.0 [110/2] via 173.17.38.1, 06:37:10, FastEthernet0/0**
   [110/2] via 173.17.39.1, 06:37:10, FastEthernet1/0

The connection of " ABR 3 " to subnet " 173.17.37.0\24 " is achieved with the routing protocol " OSPF " via either interface f 0/0 (173.17.38.2/24) to interface f 0/1 of " R 1" (173.17.38.1/24) or through the f 1/0 interface (173.17.39.1\24") to the f 1/0 interface of " ABR 2" (173.17.39.1\24").

5. **O 174.18.31.0 [110/2] via 174.18.30.2, 06:37:25, FastEthernet2/0**

The connection of " ABR 3 " to the subnet " 174.18.31.0\24 " is achieved with the routing protocol " OSPF " through interface f 2/0 (174.18.30.1/24) to interface f 0/0 of " R 5 » (174.18.30.2/24).

6. **O IA 172.16.32.0 [110/3] via 173.17.39.1, 00:59:36, FastEthernet1/0**

The connection of " ASBR 3" to the subnet "172.16.32.0\23", where all the routes of area 10 have been summarized as one " IP " address from the neighboring " Area Border Router » " ABR 2" (IA) is achieved with the routing protocol " OSPF " (O IA) from interface f 1/0 (173.17.39.2/24) to interface f 1/0 of " ABR 2" (173.17.39.1/24).